β-Kirchhoff Šoltés graphs: every rational proportion below 2/3, and a 2/3 barrier for splice constructions

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Solver: Claude Opus 5.5 (effort xhigh), with a Claude Opus 5.5 sub-agent for the literature search · Verification: pending

Abstract. A vertex v of a connected graph G is good if $G-v$ is connected and $\mathrm{Kf}(G-v) = \mathrm{Kf}(G)$, where Kf is the Kirchhoff index. Gottwald, Majstorović Ergotić and Došlić (Appl. Math. Comput. 2026) asked, for a fixed rational $\beta \in (0,1]$, for an infinite series of graphs in which at least a proportion β of the vertices are good (their Problem 4). Their splices of $K_4 - e$ have proportions $2k/(3k+1) \to 2/3$, which already answers Problem 4 for every $\beta < 2/3$. The range $\beta \geq 2/3$ is open, and we do not close it. We prove: (i) every rational $\beta \in (0, 2/3)$ is the exact proportion of good vertices of infinitely many graphs, and every real $\gamma \in (0, 2/3]$ is the limit of a strictly increasing sequence of such proportions; this answers their Problems 5 and 17 on $(0, 2/3]$; (ii) a splice lemma with an arbitrary rest, which says exactly when copies of a rooted gadget stay good; (iii) a theorem bounding the number of good vertices of any graph with a cut vertex whose good vertices lie in branches from a finite family $\mathcal{B}$ by $\lambda(\mathcal{B})|V(G)| + C(\mathcal{B})$, with an explicit linear-fractional constant $\lambda(\mathcal{B})$; (iv) an exact computation giving $\lambda = 2/3$ for all branches with at most 9 vertices, and single-gadget efficiency $\leq 2/3$ for all rooted graphs with at most 10 vertices. So splice constructions of this kind cannot answer Problem 4 for any $\beta > 2/3$; (v) C_5 is the only connected graph with at most 10 vertices in which at least two thirds of the vertices are good (rigorous modular upper bound plus exact checks). Finally, we give a 2-connected family, rook graphs $K_a \,\square\, K_b$ with one extra vertex per row, whose goodness equation has a real solution branch $a \sim b^3/2$ on which the proportion $b/(b+1)$ tends to 1. Exact arithmetic shows that this branch has no integer point with $b \leq 400$.

The question

K. K. Gottwald, S. Majstorović Ergotić, T. Došlić, *Šoltés problem for the Kirchhoff index of a graph*, Applied Mathematics and Computation 510 (2026) 129694 [1] (arXiv:2609.13751). We read the arXiv version in full and checked the definitions, Problems 4, 5, 17 and Sections 3–4 in the journal version; the statements below are identical in both (journal p. 3, arXiv p. 6):

> *"let $0 < \beta \leq 1$, $\beta \in \mathbb{Q}$. We say that a graph G is a β-Kirchhoff Šoltés graph if $|K(G)| \geq \beta|V(G)|$. i.e. the proportion of good vertices in G is at least β. We pose the following problems. Problem 4. For a fixed rational number $\beta \in (0,1]$ construct an infinite series of β-Kirchhoff Šoltés graphs."*

Here $K(G) = \{v : \mathrm{Kf}(G) = \mathrm{Kf}(G-v)\}$, and, as in [1] ("$x$ is also not a good vertex since it is a cut vertex"), a cut vertex is never good. Problem 5 asks, for a given real $\gamma \in (0,1]$, for an infinite series whose proportion of good vertices *increases* and tends to γ. Problem 17 (concluding remarks) asks: "Which (rational or irrational) numbers $\gamma \in (0,1]$ appear either as exact or as limiting values of proportions of good vertices".

The case $\beta = 1$ of Problem 4 is the Kirchhoff Šoltés problem itself (Problem 3; only C_5 is known), which we do not address [2].

What the source already gives. Theorem 15 of [1] shows that the splice $k \cdot G_1$ of k copies of $G_1 = K_4 - e$ at a vertex of degree 2 has exactly $2k$ good vertices among $3k+1$. Since $2k/(3k+1)$ increases to $2/3$, for every rational $\beta < 2/3$ the graphs $k \cdot G_1$ with $2k/(3k+1) \geq \beta$ form an infinite series of β-Kirchhoff Šoltés graphs. Problem 4 is therefore settled for $\beta < 2/3$ by the source itself, and open for $\beta \in [2/3, 1)$. The source states that $2/3$ is "the highest known limiting value of the proportion of good vertices". Its other infinite families have proportion exactly $1/2$ ($K_s + K_{s+1,1}$, Theorem 11), proportions $k(s+1)/(k(2s+1)+1) \to (s+1)/(2s+1)$ (Theorem 15) and proportions in $[1/7, 1/5)$ tending to $3 - 2\sqrt{2}$ (Theorem 10). The follow-up [3] (read in full) constructs unicyclic and cactus graphs with one or two good vertices only. For the Wiener index, the original Šoltés problem [4] and its relaxations [5] [6] [7], the best infinite families have proportions tending to $1/2$ [8] [9]; see also [10].

Results. Theorem 5 and Corollary 6 (exact proportions, Problems 5 and 17 on $(0, 2/3]$), Lemma 1, Proposition 3 (splices with an arbitrary rest), Theorem 8 and Proposition 10 (the $2/3$ barrier for splice constructions), Proposition 7 (small graphs), Proposition 13 (a 2-connected near miss). Problem 4 for $\beta \in [2/3, 1)$ remains open (Section 6).

1 Notation and the splice lemma

All graphs are finite, simple and connected. $r_G(u, w)$ is the effective resistance between u and w when every edge is a unit resistor, $\mathrm{Rt}_G(u) = \sum_w r_G(u, w)$ and $\mathrm{Kf}(G) = \sum_{\{u,w\}} r_G(u, w) = \frac{1}{2}\sum_u \mathrm{Rt}_G(u)$ [11] [12]. We write $\Delta_G(v) = \mathrm{Kf}(G) - \mathrm{Kf}(G - v)$ when $G - v$ is connected; v is good iff $\Delta_G(v) = 0$.

A *rooted graph* (B, x) is a graph with a distinguished vertex x. We put $m_B = |V(B)| - 1$ and $R_B = \mathrm{Rt}_B(x)$. A vertex $v \neq x$ is *admissible* if $B - v$ is connected; for such v let

$$\delta_B(v) = \mathrm{Kf}(B) - \mathrm{Kf}(B - v), \qquad \rho_B(v) = \mathrm{Rt}_{B-v}(x) - \mathrm{Rt}_B(x), \qquad a_B(v) = R_B - m_B\,\rho_B(v).$$

A *class* of (B, x) is a maximal set of admissible vertices with the same pair (ρ_B, δ_B). The *splice* $B \oplus W$ of two rooted graphs identifies their roots; $k \cdot B$ is the splice of k copies of B. For a real ρ and a rooted graph H put $\Phi_\rho(H) = R_H - m_H\rho$. Then m and R are additive under splicing, so Φ_ρ is too. If G has a cut vertex x, a *branch* of G at x is the subgraph induced by x and one component of $G - x$, rooted at x; G is the splice of its branches.

Lemma 1 (splice lemma). Let $G = B \oplus W$, $M = m_W$, $T = R_W$. Then $\mathrm{Kf}(G) = \mathrm{Kf}(B) + \mathrm{Kf}(W) + M\,R_B + m_B\,T$, and for every admissible $v \in V(B)$

$$\Delta_G(v) = \delta_B(v) - M\,\rho_B(v) + T.$$

A vertex $v \neq x$ of B that is not admissible is a cut vertex of G, hence not good.

Proof. Because x is a cut vertex, resistances inside B (resp. W) are those of B (resp. W), and $r_G(u, w) = r_B(u, x) + r_W(x, w)$ for $u \in V(B)$, $w \in V(W)$ (series connection through x). Summing over the pairs $u \in V(B)\backslash x$, $w \in V(W)\backslash x$ gives $MR_B + m_BT$: this is formula (9) of [1]. Since $G - v = (B - v) \oplus W$ with $m_{B-v} = m_B - 1$ and

$\mathrm{Rt}_{B-v}(x) = R_B + \rho_B(v)$, subtracting the two instances of the formula gives $\Delta_G(v) = \delta_B(v) + M(R_B - \mathrm{Rt}_{B-v}(x)) + T$. □

Proposition 13 of [1] is the special case $\delta_B(v) = 0$, and Theorem 16 of [3] is the case $B = C_5$. The rest W acts only through the two numbers (m_W, R_W).

Corollary 2 (copies). Let $G = k \cdot B \oplus W$ and let v be an admissible vertex of one copy of B, with δ, ρ, a as above. Then

$$\Delta_G(v) = \delta + (k-1)\, a + \Phi_\rho(W).$$

In particular, if $\delta = a = 0$ the copies of v are good in $k \cdot B$ for every k (Proposition 14 of [1]).

Proof. The rest of the copy containing v is $(k-1) \cdot B \oplus W$, with $M = (k-1)m_B + m_W$ and $T = (k-1)R_B + R_W$; insert this in Lemma 1. □

Proposition 3 (three regimes). Fix (B, x) and an admissible v with δ, ρ, a.

(1) If $\rho \le 0$, then $a \ge R_B > 0$, and for $k > 1 - \delta/a$ no copy of v is good in $k \cdot B \oplus W$, whatever W.

(2) If $\rho > 0$ and a copy of v is good in $k \cdot B \oplus W$, then $m_W \ge (\delta + (k-1)a)/\rho$.

(3) Suppose $a < 0$. Let P_L be a path with L edges rooted at an end, so $\Phi_\rho(P_L) = L(L+1)/2 - L\rho$. Suppose there are a rooted graph A and an integer $L_0 \ge 0$ with $\delta + \Phi_\rho(A) + \Phi_\rho(P_{L_0}) \in |a|\mathbb{Z}$. Let $N \ge 1$ be an integer with $N\rho \in \mathbb{Z}$ and $s := N|a| \in \mathbb{Z}$, and put $L_t = L_0 + 2st$, $k_t = 1 + (\delta + \Phi_\rho(A) + \Phi_\rho(P_{L_t}))/|a|$. Then $k_t \in \mathbb{Z}$ and $k_t \ge 1$ for t large. In $G_t = k_t \cdot B \oplus P_{L_t} \oplus A$ every vertex of every copy of B with the pair (ρ, δ) is good. If g vertices of B have this pair, then $|K(G_t)|/|V(G_t)| \ge k_t g/(1 + k_t m_B + L_t + m_A) \to g/m_B$.

Proof. (1) $a = R_B - m_B\rho \ge R_B > 0$, and $\Phi_\rho(W) = R_W - m_W\rho \ge 0$, so $\Delta \ge \delta + (k-1)a > 0$ by Corollary 2. (2) $\Delta = 0$ and $R_W \ge 0$ give $m_W\rho = \delta + (k-1)a + R_W \ge \delta + (k-1)a$. (3) $N(\Phi_\rho(P_{L+2s}) - \Phi_\rho(P_L)) = Ns(2L+2s+1) - 2s\, N\rho \in s\mathbb{Z}$, so $\Phi_\rho(P_{L_t}) - \Phi_\rho(P_{L_0}) \in |a|\mathbb{Z}$ and $k_t \in \mathbb{Z}$; $k_t \to \infty$ because $\Phi_\rho(P_L) \sim L^2/2$. By Corollary 2 with $W = P_{L_t} \oplus A$, $\Delta = \delta - (k_t - 1)|a| + \Phi_\rho(A) + \Phi_\rho(P_{L_t}) = 0$, and the equation depends on v only through (ρ, δ). Finally $L_t = O(\sqrt{k_t})$. □

So a gadget class with $a = 0$ (the case of [1]) or $a < 0$ yields families whose proportion of good vertices tends to g/m_B; a class with $a > 0$ needs m_W at least linear in k.

Example 4 (C_5, the regime $a < 0$). Root C_5 at a vertex. Its two neighbours have $(\delta, \rho, a) = (0, 2, -4)$; the two other vertices have $(0, 0, 4)$. With $A = \varnothing$, $L_0 = 3$ ($\Phi_2(P_3) = 6 - 6 = 0$), $N = 1$, $s = 4$: $k = 1 + L(L-3)/8$ for $L \equiv 3 \pmod 8$; also $L \equiv 0 \pmod 8$ works. The graphs $k \cdot C_5 \oplus P_L$ with $(L, k) = (3, 1), (8, 6), (11, 12)$ have exactly $2k$ good vertices: proportions $1/4$, $4/11$, $2/5$, tending to $1/2$. The first one is the graph $C_5 v_5 P_4$ of [3]. (Checked exactly by `check_splice_lemma.py`.) ⌟

2 Every rational proportion below 2/3

Let D be $K_4 - e$ rooted at a vertex x of degree 2 (vertices x, y of degree 2, a, b of degree 3). Let D' be $K_4 - e$ rooted at a vertex of degree 3, E an edge rooted at an end, and K_4 rooted at any vertex. Put

$$F_5 = D' \oplus E \oplus E, \qquad F_6 = K_4 \oplus E \oplus E \oplus E, \qquad G(k,\alpha,\gamma) = k \cdot D \oplus \alpha \cdot F_5 \oplus \gamma \cdot F_6,$$

so that $|V(G(k,\alpha,\gamma))| = n = 1 + 3k + 5\alpha + 6\gamma$.

Theorem 5. For every $k \geq 1$ and $\alpha, \gamma \geq 0$, the good vertices of $G(k,\alpha,\gamma)$ are exactly the $2k$ vertices of degree 3 of the copies of D. Consequently, for every rational $\beta = p/q \in (0, 2/3)$ in lowest terms and every integer j with $(2q - 3p)j \geq 21$, there are $\alpha, \gamma \geq 0$ with $5\alpha + 6\gamma = (2q - 3p)j - 1$, and $G(pj, \alpha, \gamma)$ has $n = 2qj$ vertices of which exactly βn are good. In particular every rational $\beta \in (0, 2/3)$ is the exact proportion of good vertices of infinitely many graphs.

Proof. *Constants.* Series–parallel reduction gives, in D: $r(x,a) = r(x,b) = 5/8$, $r(x,y) = 1$, so $R_D = 9/4$; $D - a$ is the path $x\,b\,y$ ($\mathrm{Rt}(x) = 3$); $D - y$ is a triangle ($\mathrm{Rt}(x) = 4/3$). In D' rooted at r, the other degree-3 vertex b' has $r(r,b') = 1/2$, the degree-2 vertices u, w have $r(r,u) = r(r,w) = 5/8$, so $R_{D'} = 7/4$; $D' - b'$ is the path $u\,r\,w$ ($\mathrm{Rt}(r) = 2$) and $D' - u$ a triangle. With $\mathrm{Kf}(K_4 - e) = 4$, $\mathrm{Kf}(P_3) = 4$, $\mathrm{Kf}(K_3) = 2$, $\mathrm{Kf}(K_4) = 3$, $\mathrm{Kf}(K_2) = 1$ we get:

branch, vertex	R	δ	ρ	a
D: a, b	9/4	0	3/4	0
D: y	9/4	2	−11/12	5
D': b'	7/4	0	1/4	1
D': u, w	7/4	2	−5/12	3
K_4: any	3/2	1	−1/6	2
E: leaf	1	1	−1	2

Balance at the hub. $R_D = \frac{3}{4} \cdot 3$, $R_{F_5} = \frac{7}{4} + 2 = \frac{3}{4} \cdot 5$ and $R_{F_6} = \frac{3}{2} + 3 = \frac{3}{4} \cdot 6$, hence $\mathrm{Rt}_G(x) = \frac{3}{4}(n-1)$. Let $v \neq x$ lie in a branch H of G at x (a copy of D, D', K_4 or E). The rest of H has $M = n - 1 - m_H$ and $T = \frac{3}{4}(n-1) - R_H$, so Lemma 1 gives

$$\Delta_G(v) = \delta_H(v) + (n-1)\left(\tfrac{3}{4} - \rho_H(v)\right) - a_H(v).$$

From the table, $\Delta_G = 0$ for a, b in the copies of D, and $\Delta_G = \frac{5}{3}(n-1) - 3$, $\frac{1}{2}(n-1) - 1$, $\frac{7}{6}(n-1) - 1$, $\frac{11}{12}(n-1) - 1$, $\frac{7}{4}(n-1) - 1$ for the vertices y, b', u, w, the vertices of K_4, and the leaves. These are positive as soon as $n \geq 4$ ($n \geq 9$ when D' occurs). The root x is a cut vertex as soon as G has two branches; for $G(1,0,0) = K_4 - e$ one has $\Delta(x) = \Delta(y) = \mathrm{Kf}(K_4 - e) - \mathrm{Kf}(K_3) = 2$.

Counting. With $k = pj$, the equation $n = 2qj$ reads $5\alpha + 6\gamma = (2q - 3p)j - 1$, and $2q - 3p \geq 1$. By Sylvester's theorem on the Frobenius number, every integer $\geq (5-1)(6-1) = 20$ is of the form $5\alpha + 6\gamma$ with $\alpha, \gamma \geq 0$. Then $|K(G)| = 2pj = \beta n$. Different j give different orders. □

Corollary 6. **(1)** For every rational $\beta \in (0, 2/3)$ there are infinitely many graphs in which exactly $\beta|V(G)|$ vertices are good.

(2) (Problem 5 for $\gamma \le 2/3$.) For every real $\gamma \in (0, 2/3]$ there are graphs $G_1, G_2, \ldots$ of strictly increasing orders whose proportions of good vertices increase strictly and tend to γ.

(3) (Problem 17 on $(0, 2/3]$.) Every rational number in $(0, 2/3)$ is an exact proportion of good vertices, each for infinitely many graphs; every real number in $(0, 2/3]$ is a limiting proportion; and 1 is an exact proportion (C_5).

Proof. (1) is Theorem 5. For (2) choose rationals $\beta_1 < \beta_2 < \ldots$ in $(0, 2/3)$ tending to γ, and inductively a graph G_i with proportion exactly β_i and $|V(G_i)| > |V(G_{i-1})|$. (3) follows. □

For $\beta \ge 2/3$ the corollary says nothing. Whether any rational in $[2/3, 1)$ is an exact proportion is open; Proposition 7 shows that no graph with $6 \le n \le 10$ vertices has proportion $\ge 2/3$.

3 Small graphs

Proposition 7 (computer-assisted). For $n = 3, 4, \ldots, 10$, the largest number of good vertices of a connected graph on n vertices is $0, 2, 5, 3, 4, 4, 4, 6$. For $n = 4, \ldots, 10$ the maximum is attained by exactly one graph, except for $n = 8$ (two graphs): $K_4 - e$; C_5; $K_2 + K_{3,1}$; $2 \cdot (K_4 - e)$; $K_3 + K_{4,1}$ and the graph with graph6 code `G?qa‘o`; the graph `H??F?zW`; $3 \cdot (K_4 - e)$. Here $k \cdot (K_4 - e)$ is the splice of k copies at a vertex of degree 2. In particular C_5 is the only connected graph with at most 10 vertices in which at least two thirds of the vertices are good; for $6 \le n \le 10$ the proportion is at most $3/5$.

Proof (Method). For a connected graph H with grounded Laplacian A, $\tau(H) = \det A$ and $\mathrm{Kf}(H) = N \operatorname{tr} A^{-1} - \mathbf{1}^{\mathsf{T}} A^{-1} \mathbf{1} = \alpha(H)/\tau(H)$ with $\alpha(H) \in \mathbb{Z}$. Modulo a prime $p \nmid \tau(H)$ we compute $k_p(H) = \alpha(H)\tau(H)^{-1} \bmod p$. If $k_p(G) \ne k_p(G - v)$, then $\alpha(G)\tau(G - v) - \alpha(G - v)\tau(G)$ is not divisible by p, so v is not good. The program `goodcount.c` counts the vertices that survive this test for two primes $p < 2^{30}$ (a test is skipped when $p \mid \tau$); the count $\mathrm{pg}(G)$ is a rigorous upper bound for $|K(G)|$. It was run on all 11 989 762 connected graphs with $3 \le n \le 10$ (nauty `geng -c` [13]; the counts agree with OEIS A001349 [14]). Every graph with $\mathrm{pg} \ge 2n/5$ was then checked with exact rationals (`verify_small_graphs.py`): pg equals the exact count for all of them, and $\mathrm{pg} \ge 2n/3$ occurs only for C_5. Validation: $\mathrm{pg} = |K(G)|$ for all 12 111 connected graphs with $n \le 8$ (`test_c_programs.py`). □

4 A 2/3 barrier for splice constructions

All infinite families of [1] [3] with good vertices in a positive proportion are splices at one vertex. The next theorem bounds what any such construction can give, in terms of the branches that contain good vertices.

For a finite set $\mathcal{B}$ of rooted graphs and $\hat{\rho} > 0$, let $\mathcal{C}_{\mathcal{B}}(\hat{\rho})$ be the set of triples (g, m, a)

such that some $(B,x)\in\mathcal{B}$ has a class O with $\rho_B=\hat\rho$ on O, $g=|O|$, $m=m_B$ and $a=R_B-m_B\hat\rho$. Put

$$\sigma(g,m,a)=\begin{cases} g/m, & a\le 0,\\ g\hat\rho/(a+m\hat\rho), & a>0,\end{cases}\qquad \pi((g_1,m_1,a_1),(g_2,m_2,a_2))=\frac{|a_2|g_1+a_1g_2}{|a_2|m_1+a_1m_2}\quad (a_1>0>a_2),$$

and let $\lambda(\mathcal{B})$ be the maximum, over $\hat\rho>0$, of all values σ on $\mathcal{C}_{\mathcal{B}}(\hat\rho)$ and π on pairs from $\mathcal{C}_{\mathcal{B}}(\hat\rho)$ with $a_1>0>a_2$ (and $\lambda=0$ if no class has $\rho>0$). For a single rooted graph, $\sigma=(g/m)\min(1,\hat\rho m/R_B)$ is the *efficiency* of the class.

Theorem 8. Let $\mathcal{B}$ be a finite set of rooted graphs. There is a constant $C(\mathcal{B})$ such that every connected graph G with a cut vertex x satisfies $|K(G)|\le\lambda(\mathcal{B})\,|V(G)|+C(\mathcal{B})$ if every good vertex of G lies in a branch of G at x that is isomorphic, as a rooted graph, to a member of $\mathcal{B}$.

Proof. Let $n=|V(G)|$, let $H_1,\dots,H_r$ be the branches at x, $m_i=m_{H_i}$, $R_i=R_{H_i}$, and $T=\mathrm{Rt}_G(x)=\sum_i R_i$. By Lemma 1 with W the union of the other branches, a vertex v of H_i is good iff it is admissible in H_i and

$$T=(n-1-m_i)\,\rho_i(v)+R_i-\delta_i(v). \tag{9}$$

Step 1 (one value of ρ). Only finitely many values of (m,R,ρ,δ) occur in $\mathcal{B}$. If $v\in H_i$ and $w\in H_j$ are good with $\rho_i(v)\ne\rho_j(w)$, subtracting the two instances of (9) gives $n-1\le\Lambda(\mathcal{B})$, a constant. So if $n-1>\Lambda(\mathcal{B})$, all good vertices have the same $\rho=\hat\rho$, and by (9) the same $c^*=R_i-\delta_i(v)-m_i\hat\rho$, with $T=(n-1)\hat\rho+c^*$. In particular, in each branch the good vertices form one class.

Step 2 ($\hat\rho\le 0$). Then $a_i=R_i-m_i\hat\rho\ge R_i$ for every branch, and $\sum_i a_i=T-(n-1)\hat\rho=c^*$ is bounded. Each branch containing a good vertex has $R_i\ge\min_{\mathcal{B}}R>0$, so there are boundedly many such branches and $|K(G)|$ is bounded.

Step 3 ($\hat\rho>0$). Let I be the set of branches containing good vertices; branch $i\in I$ carries a triple $(g_i,m_i,a_i)\in\mathcal{C}_{\mathcal{B}}(\hat\rho)$ with $a_i=R_i-m_i\hat\rho$, and $|K(G)|=\sum_{i\in I}g_i$. Every branch satisfies $a_j=R_j-m_j\hat\rho>-m_j\hat\rho$, and $\sum_j a_j=c^*$ over all branches. Hence $\sum_{j\notin I}m_j>(\sum_{i\in I}a_i-c^*)/\hat\rho$, and

$$n-1\ \ge\ D-|c^*|/\hat\rho,\qquad D:=\sum_{i\in I}m_i+\max\Big(0,\tfrac{1}{\hat\rho}\sum_{i\in I}a_i\Big).$$

Now $\sum_i g_i\le\lambda D$. Indeed, grouping equal triples with weights $x_t\ge 0$, the ratio $\sum x_tg_t/D(x)$ is linear-fractional on each of the two cones $\sum x_ta_t\le 0$ and $\sum x_ta_t\ge 0$. On the first, normalise $\sum x_tm_t=1$: the maximum of the linear function $\sum x_tg_t$ is attained at a vertex, i.e. a single triple with $a\le 0$ (value g/m) or a pair with $a_1>0>a_2$ on the hyperplane $\sum x_ta_t=0$ (value π). On the second, $D=\sum x_tR_t/\hat\rho$; normalise it to 1 and conclude in the same way (single triples with $a\ge 0$, value $g\hat\rho/R$, or the same pairs). Therefore $|K(G)|\le\lambda(n-1+|c^*|/\hat\rho)$. There are finitely many possible $(\hat\rho,c^*)$; take $C(\mathcal{B})$ larger than $\Lambda(\mathcal{B})+1$, than $\lambda|c^*|/\hat\rho$, and than the bound of Step 2. □

Proposition 10 (exact computation)**.** Let $\mathcal{B}_N$ be the set of all rooted connected graphs with at most N vertices. Then:

(1) every class of every rooted graph with at most 10 vertices has efficiency $\sigma \le 2/3$. Equality occurs only on 4, 7 and 10 vertices (2, 5 and 11 rooted classes, all with $g/m = 2/3$ and $a \le 0$); examples are the two degree-3 vertices of $K_4 - e$ rooted at a degree-2 vertex and the degree-3 vertices of $2 \cdot (K_4 - e)$ and $3 \cdot (K_4 - e)$. Among classes with $g/m > 2/3$ the largest efficiency is 0.469 (on 9 vertices; 0.164 on 10 vertices), so their ratio ρ/ρ^* is below 0.63;

(2) $\lambda(\mathcal{B}_9) = 2/3$ (8 915 804 classes with $\rho > 0$): no balanced pair improves on single classes. More precisely, no class with $g/m > 2/3$ and $a > 0$ shares its value of ρ with a class with $a < 0$, and a pair value is a weighted mediant of g_1/m_1 and $g_2/m_2 = \sigma_2 \le 2/3$.

Proof (Method). `gadgetbound.c` computes, for every connected graph B (nauty) and every root, all quantities in exact integer arithmetic: τ and $\operatorname{adj} A$ by fraction-free Gauss–Jordan elimination [15] in 128-bit integers, then $\tau\, r(u,w)$, $\tau\,\mathrm{Rt}$, $\tau\,\mathrm{Kf}$, the classes by cross-multiplication, and the test $\sigma \le 2/3$ by cross-multiplication. Every pivot is a leading principal minor of a positive definite matrix, and every division is checked to be exact. `check_mixing_bound.py` rebuilds the classes as exact fractions from the program's output and evaluates all pairs. Validation against a separate exact implementation (python-flint) on all rooted graphs with $n \le 7$: `test_c_programs.py`. □

Corollary 11. There is a constant C such that every graph G with a cut vertex x, whose good vertices all lie in branches at x with at most 9 vertices, has $|K(G)| \le \frac{2}{3}|V(G)| + C$. For a single rooted graph B with at most 10 vertices, $\limsup_{k\to\infty} \sup_W \; |K(k \cdot B \oplus W) \cap (\text{copies of } B)| \,/\, |V(k \cdot B \oplus W)| \le 2/3$. So constructions of this type (in particular all those of [1] [3]) cannot give an infinite series of β-Kirchhoff Šoltés graphs with $\beta > 2/3$.

Proof. The first claim is Theorem 8 with Proposition 10(2). For the second, apply Step 1 of the proof of Theorem 8 to the branches that are copies of B: for large k the good vertices in the copies form one class. Its contribution is bounded by its efficiency, by Proposition 3(1)–(2) and $|V| \ge 1 + km_B + m_W$. Proposition 10(1) gives $\sigma \le 2/3$. □

Remark 12 (other structures explored, floating point only). We also explored, in floating point and without finding any proportion above 1/2 for an exact solution:

- cographs with at most four twin classes of sizes ≤ 14 (maximum 1/2);
- blow-ups of all connected patterns with at most four vertices into cliques or independent sets of sizes ≤ 20 (no proportion ≥ 0.55);
- theta graphs with threads of lengths 1–4;
- rook-type graphs with row or column hubs, hub cliques and super-hubs on a 9×9 grid of sizes;
- joins $K_r + H$ and $\overline{K_r} + H$ with H vertex-transitive. Here the H-class is good only at a non-integral $r^* < 1$ for sparse H (cycles C_m, $m \ge 6$, the dodecahedron): a hub coupled to every vertex of H is too strong.

Scripts and logs are in `scripts/exploration/` and `certificates/exploration/`.

5 A 2-connected near miss

The barrier above concerns cut vertices. Let $R(a,b)$ be the rook graph $K_a \,\square\, K_b$ (vertices (i,j), rows and columns are cliques) with one extra vertex h_i per row adjacent to the b vertices of row i. It has $n = a(b+1)$ vertices, it is 2-connected, and its ab rook vertices form one orbit of $S_a \times S_b$. If they are good, the proportion is $b/(b+1)$.

Proposition 13. **(1)** The Kirchhoff index is

$$\mathrm{Kf}(R(a,b)) = ab + \frac{(a-1)(a+b+1)(b+1)}{b} + \frac{a(a-1)(b-1)(b+1)}{a+b+1}\,.$$

(2) For a rook vertex v, $\mathrm{Kf}(R)-\mathrm{Kf}(R-v) = -P(a,b)/\big[b(b-1)(b+1)(a+b+1)(a^2b+ab^2-b-1)(ab+a+b^2+b-1)\big]$, where $P \in \mathbb{Z}[a,b]$ is the irreducible polynomial of degree 5 in a and 8 in b printed in `certificates/check_rook_hubs.log`; its terms of largest weight for $a \asymp b^3$ are $a^4b^3(2a-b^3)$. The denominator is positive for $a,b \geq 2$.

(3) For every $3 \leq b \leq 400$, $P(\cdot,b)$ has exactly one real root $a^*(b) > 1.5$, and the sign of $\mathrm{Kf}(R) - \mathrm{Kf}(R-v)$ changes between $\lfloor a^* \rfloor$ and $\lceil a^* \rceil$. So the rook class is good along a real curve on which the proportion $b/(b+1)$ tends to 1. Numerically $a^*(b)/b^3 \to 1/2$ (0.4987 at $b = 400$).

(4) There is no integer $a \geq 2$ with $P(a,b) = 0$ for $2 \leq b \leq 400$: no graph $R(a,b)$ with $b \leq 400$ has good rook vertices.

(5) With c independent hubs per row (graph $R_c(a,b)$, proportion $b/(b+c)$), the same method gives $\mathrm{Kf}(R_c)-\mathrm{Kf}(R_c-v)$ as an explicit rational function, of degree 5 in a in the numerator. For $c < b/2$ it has real zeros: for instance $a^* \approx 61.83$ for $(b,c) = (10,3)$ (proportion $10/13$), and $a^* \approx 2.04$ for $(b,c) = (6,3)$. But there is no integer solution $a \geq 2$ with $3 \leq b \leq 200$ and $1 \leq c \leq b$.

Proof. (1)–(2): $\mathbb{R}^V$ decomposes under $S_a \times S_b$ into isotypic components, on which the Laplacian acts by small matrices. For R: $\{0, b+1\}$ on the trivial part, $\left(\begin{smallmatrix} a+1 & -1 \\ -b & b \end{smallmatrix}\right)$ with multiplicity $a-1$, $b+1$ with multiplicity $b-1$, $a+b+1$ with multiplicity $(a-1)(b-1)$. For $R-v$, $v=(1,1)$, under $S_{a-1} \times S_{b-1}$, the cells are: row-mates, column-mates, other rook vertices, the hub of row 1, and the other hubs. This gives a 5×5 quotient Q with a simple zero eigenvalue, a 3×3 block with multiplicity $a-2$, a 2×2 block with multiplicity $b-2$, and the eigenvalue $a+b+1$ with multiplicity $(a-2)(b-2)$; the blocks are written in `check_rook_hubs.py`. Then $\mathrm{Kf} = n \operatorname{tr} L^+$, and the trace over a block is a ratio of characteristic-polynomial coefficients. The symbolic results agree with exact brute-force values for 12 pairs (a,b). (3)–(4) are exact computations on the integer polynomial P: real-root isolation, exact evaluation at $\lfloor a^* \rfloor, \lceil a^* \rceil$, and factorisation over $\mathbb{Q}$ of $P(\cdot,b)$ to exclude integer roots. (5): the hubs of a row are false twins. They contribute the eigenvalue b (multiplicity $a(c-1)$) in R_c, and $b-1$ (multiplicity $c-1$) and b (multiplicity $(a-1)(c-1)$) in $R_c - v$; the other blocks are those above with the hub cells weighted by c. The formula agrees with brute force for 8 triples (a,b,c) and

reduces to (1)–(2) for $c = 1$ (`check_rook_hubs_c.py`). □

Along the branch $a \approx b^3/2$ the Puiseux expansion of $a^*(b)$ does not terminate (P is irreducible), so one expects only finitely many integer points. An infinite family would need one more integer parameter entering the goodness equation in a solvable way (linearly, or through a Pell equation as in Theorem 10 of [1]). In our tests, the natural candidates (several hubs per row, hubs on rows and columns, rows joined through column hubs, universal vertices) gave real solution curves with proportions above 2/3, but no integer solution.

6 Open problems and limitations

- Problem 4 is open for every rational $\beta \in [2/3, 1)$. Even $\beta = 2/3$ is open: no known infinite series has at least two thirds of its vertices good. By Proposition 7, no graph with $6 \le n \le 10$ has proportion $\ge 2/3$.
- Corollary 11 covers splices whose good vertices lie in branches with at most 9 vertices (mixtures of branch types), and single gadget types with at most 10 vertices. Larger branches are not excluded. However, any improvement over 2/3 by a splice needs a rooted graph with a class of efficiency $> 2/3$, and in all rooted graphs with at most 10 vertices the classes covering more than two thirds of the non-root vertices have $\rho/\rho^* < 0.63$ ($\rho^* = R_B/m_B$).
- The real branch of Proposition 13 shows that 2-connected product-like graphs can come arbitrarily close to proportion 1 in the real relaxation. Turning this into integer solutions is the most promising route we found.
- Floating-point computations (the explorations of Section 4 and the ratio $a^*(b)/b^3$) are evidence, not proofs. Every claim stated in a theorem, proposition or corollary rests on exact rational or modular arithmetic.

7 Computer verification

Environment: Fedora 44 under WSL, Python 3.12, python-flint 0.6, sympy 1.14, networkx 3, nauty 2.8.9, gcc. From the folder of this problem:

script	claims (PASS/FAIL, exit code)	time
`check_splice_lemma.py`	Lemma 1, Corollary 2 on random instances, Example 4	3 s
`check_exact_proportions.py`	constants and neutrality of Theorem 5; good sets of all 224 graphs $G(k, \alpha, \gamma)$ with $n \le 46$; arithmetic for 11 values of β; six graphs with prescribed β	15 s
`check_rook_hubs.py`	Proposition 13(1)–(4)	45 s
`check_rook_hubs_c.py`	Proposition 13(5) (argument 200; 60 takes 75 s)	15 min
`goodcount.c`, `run_goodcount.sh`, `verify_small_graphs.py`	Proposition 7	5 min
`gadgetbound.c`, `check_mixing_bound.py`	Proposition 10 (`gadgetbound` for $N \le 10$: 50 min; `check_mixing_bound.py 9`: 25 min)	75 min
`test_c_programs.py`	both C programs against exact Python on all graphs with $n \le 8$ ($n \le 7$ rooted)	30 s

Logs are in `certificates/` and `certificates/search/`. The runs of `gadgetbound` for $N = 3, \ldots, 10$ report no class with $\sigma > 2/3$ (files `gadgetbound_n*.log`).

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